100 Practice Exam Questions and Answers

Databricks Certified **Machine Learning Associate**

Topics: Feature Store (Unity Catalog), MLflow, AutoML, Data Processing, Hyperopt, Spark ML, Model Evaluation, Model Serving

Practice Exam

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Answer Key (Compact) **20**

Exam Format & How to Use

The Databricks ML Associate exam assesses the ability to **use Databricks** for basic ML tasks: AutoML, Unity Catalog, MLflow, feature engineering, model building (training/tuning/evaluation), and deployment [citation:1]. It is **not** a pure math or ML theory exam. All ML code is Python; data manipulation may appear in SQL.

- **[MC]** = multiple choice (single best answer).
- **[SA]** = short answer.
- **[CODE]** = write or interpret Databricks/MLflow code.
- **[SCENARIO]** = scenario-based reasoning.

Domains (official weights): A. Databricks Machine Learning (38%); B. ML Workflows (19%); C. Model Development (31%); D. Model Deployment (12%) [citation:1].

1 A. Databricks Machine Learning (38%)

Feature Store & Unity Catalog

Q1. **[MC]** What is the primary advantage of creating Feature Store tables at the **account level** in Unity Catalog versus the workspace level?

(a) Faster training speed

(b) Cross-workspace governance, discoverability, and access control

(c) Automatic feature engineering

(d) Lower storage cost

Answer: (b). Account-level tables in Unity Catalog are governed and discoverable across workspaces [citation:1].

Q2. **[SA]** Which client is used to create and manage Feature Store tables in Unity Catalog?

Answer: `FeatureEngineeringClient`.

Q3. **[SA]** What method creates a feature table?

Answer: `fe.create_table()`.

Q4. **[SA]** What method writes features to a feature table?

Answer: `fe.write_table()`.

Q5. **[SA]** What does `fe.create_training_set()` do that a plain join does not?

Answer: It performs **point-in-time lookups**, ensuring training data uses only feature values that were available at the time of each label event [citation:2].

Q6. **[SCENARIO]** A team's offline model scores well but production predictions degrade. Review shows a feature was computed using data only available *after* the label outcome. What is the diagnosis?

Answer: Training-serving skew / target leakage. Point-in-time lookups in Feature Store prevent this by aligning feature values to the label timestamp [citation:2].

Q7. **[MC]** What is the difference between online and offline feature tables?

(a) Online is for training; offline is for serving

(b) Online serves low-latency real-time inference; offline is for batch training/scoring

(c) They are identical

(d) Offline is deprecated

Answer: (b).

MLflow

Q8. **[SA]** Which MLflow API call manually logs a metric?

Answer: `mlflow.log_metric()`.

Q9. **[SA]** Which MLflow API call registers a model in Unity Catalog?

Answer: `mlflow.register_model()`.

Q10. **[MC]** What replaced the old "Staging" and "Production" stages for model promotion in Unity Catalog?

(a) Tags only

(b) Aliases (e.g., `@champion`)

(c) Notebook revisions

(d) Git branches

Answer: (b). Aliases allow applications to reference `@champion`, and promotion is an alias move [citation:2].

Q11. **[SA]** How do you promote a challenger model to champion using aliases?

Answer: Move the `champion` alias to the new model version.

Q12. **[SCENARIO]** A team needs every training run to be reproducible months later, including parameters, metrics, and the model artifact. What is the correct Databricks-native approach?

Answer: Use **MLflow tracking** to log parameters, metrics, artifacts, and the model for each run [citation:20].

AutoML

Q13. **[SA]** What does Databricks AutoML produce besides a trained model?

Answer: A **leaderboard** of candidate models, editable notebooks for each trial, and automatic feature engineering [citation:15].

Q14. **[SA]** Which API runs classification AutoML?

Answer: `databricks.automl.classify()`.

Q15. **[MC]** AutoML is *least* appropriate when:

(a) You need a quick baseline

(b) You need a highly customized deep learning architecture

(c) You want automatic feature engineering

(d) You want to compare multiple algorithms

Answer: (b).

2 B. ML Workflows (19%)

Data Processing & EDA

Q16. **[SA]** How do you compute summary statistics on a Spark DataFrame?

Answer: `df.summary()` or `dbutils.data.summarize()` [citation:1].

Q17. **[SA]** How do you remove outliers using IQR in Spark?

Answer: Compute Q1 and Q3, then filter values outside $[Q1 - 1.5 \cdot IQR,\ Q3 + 1.5 \cdot IQR]$ [citation:1].

Q18. **[MC]** When is median imputation preferred over mean imputation?

(a) When data is symmetric

(b) When data is skewed or contains outliers

(c) When there are no missing values

(d) Always

Answer: (b). Median is robust to skew and outliers [citation:1].

Q19. **[MC]** One-hot encoding is *least* appropriate for:

(a) Nominal categorical features

(b) Very high-cardinality categorical features

(c) Low-cardinality categorical features

(d) Tree-based models

Answer: (b). High cardinality causes dimensionality explosion [citation:1].

Q20. **[SA]** How do you one-hot encode in Spark ML?

Answer: `StringIndexer` followed by `OneHotEncoder`.

Q21. **[SA]** What does `VectorAssembler` do?

Answer: Combines multiple feature columns into a single vector column required by Spark ML estimators.

Q22. **[SA]** When is log transformation appropriate?

Answer: For **right-skewed positive** features [citation:1].

3 C. Model Development (31%)

Hyperparameter Tuning

Q23. **[SA]** What does Hyperopt's `fmin` do?

Answer: Minimizes an objective function over a defined search space [citation:15].

Q24. **[SA]** What does `SparkTrials` do?

Answer: Distributes Hyperopt trials across a Spark cluster for parallel tuning [citation:2].

Q25. **[SCENARIO]** A team has 8 worker nodes and needs to tune a model with 5 hyperparameters. Which approach best leverages the cluster?

(a) Grid search on the driver node

(b) Sequential manual tuning

(c) Hyperopt with `SparkTrials(parallelism=8)` and TPE search

(d) Random search on the driver node

Answer: (c). Bayesian (TPE) search is more efficient than grid/random, and `SparkTrials` distributes trials across workers [citation:9].

Q26. **[SA]** With 5-fold cross-validation and a grid of 10 combinations, how many models are trained?

Answer: 50 (5×10) [citation:1].

Model Evaluation

Q27. **[MC]** A fraud dataset has 1% positive class. A model reports 99% accuracy. What should you do?

(a) Deploy it

(b) Evaluate precision, recall, F1, and PR-AUC on the minority class

(c) Increase epochs

(d) Add features

Answer: (b). Accuracy is misleading; always predicting majority gives 99% [citation:2].

Q28. **[SA]** What does $R^2 = 0.85$ mean?

Answer: The model explains 85% of the variance in the target [citation:7].

Q29. **[SA]** Why might RMSE be higher than MAE?

Answer: RMSE penalizes large errors more heavily due to squaring, indicating presence of larger prediction errors [citation:7].

Q30. **[SCENARIO]** Training accuracy is 99.5%, validation is 72%. What is the diagnosis and fix?

(a) Underfitting; increase depth

(b) Overfitting; reduce complexity or add regularization

(c) Good model; deploy

(d) Need more test data

Answer: (b). Large train-validation gap signals overfitting [citation:9].

Q31. **[MC]** What is the effect of increasing model complexity on bias and variance?

(a) Bias decreases, variance increases

(b) Bias increases, variance decreases

(c) Both decrease

(d) Both increase

Answer: (a).

Q32. **[SA]** What does `CrossValidator` do in Spark ML?

Answer: Performs k-fold cross-validation with a `ParamGridBuilder` to select the best hyperparameters.

4 D. Model Deployment (12%)

Q33. **[SA]** What are the three model serving approaches tested on the exam?

Answer: Batch, **real-time**, and **streaming** [citation:1].

Q34. **[SCENARIO]** A scikit-learn model must score 100 million Delta table records efficiently. What is the best approach?

(a) Load into pandas on the driver and loop

(b) Use `mlflow.pyfunc.spark_udf()` to distribute inference across workers

(c) Export to CSV and score externally

(d) Create a real-time endpoint and call per record

Answer: (b). Spark UDF distributes inference across the cluster [citation:9].

Q35. **[SA]** How is streaming inference performed in Databricks?

Answer: Using **Delta Live Tables** [citation:1].

Q36. **[SA]** How do you split traffic between real-time serving endpoints?

Answer: Configure **traffic splitting** on the serving endpoint [citation:1].

Q37. **[MC]** What is the primary purpose of a real-time serving endpoint?

(a) Batch scoring

(b) Low-latency predictions accessible via REST API

(c) Training models

(d) Data storage

Answer: (b).

Answer Key

Q	Answer	Q	Answer
1	(b) cross-workspace governance	21	VectorAssembler combines columns
2	`FeatureEngineeringClient`	22	right-skewed positive features
3	`fe.create_table()`	23	minimizes objective function
4	`fe.write_table()`	24	distributes trials across Spark
5	point-in-time lookups	25	(c) Hyperopt + SparkTrials
6	training-serving skew / leakage	26	50 models
7	(b) online low-latency, offline batch	27	(b) precision/recall/F1/PR-AUC
8	`mlflow.log_metric()`	28	85% variance explained
9	`mlflow.register_model()`	29	RMSE penalizes large errors
10	(b) aliases (`@champion`)	30	(b) overfitting; reduce complexity
11	move champion alias to new version	31	(a) bias down, variance up
12	MLflow tracking logs params/metrics/artifacts	32	k-fold CV with ParamGridBuilder
13	leaderboard + editable notebooks + auto FE	33	batch, real-time, streaming
14	`databricks.automl.classify()`	34	(b) spark_udf distributes inference
15	(b) custom deep learning	35	Delta Live Tables
16	`df.summary()` / `dbutils.data.summarize()`	36	traffic splitting on endpoint
17	IQR filter (Q1-1.5IQR, Q3+1.5IQR)	37	(b) low-latency REST API
18	(b) skewed or outliers		
19	(b) high cardinality		
20	StringIndexer + OneHotEncoder		

Note: This set is **objective-derived practice**, not official exam questions. Databricks does not publish a question bank. Every item above maps to a published objective in the official exam guide [citation:1].